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A deep learning-based crop pest and disease recognition system for farmland in Xinjiang

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Abstract: **With the rapid development of modern agriculture, the timely detection and treatment of pests and diseases is very important to ensure the healthy growth of crops and improve production. This paper presents a deep learning-based system for identifying crop diseases and pests in Xinjiang, utilizing Convolutional Neural Networks (CNNs). The system achieves over 95% accuracy, outperforming traditional methods. It offers real-time pest control suggestions, enhancing both detection efficiency and accuracy.**

Key words: deep learning; Plant diseases and insect pests recognition; Convolution neural network (CNN); Crop health; Image processing; ResNet - 50; Agricultural science and technology

I. INTRODUCTION

As an important part of the national economy, agriculture is directly related to food security and the income of farmers. However, crops are often attacked by various diseases and pests in the growth process, which seriously affects the yield and quality of crops. Traditional pest detection methods mainly rely on manual observation and experience judgment, which are not only time-consuming and labor-intensive, but also difficult to ensure the accuracy and timeliness. With the rapid development of artificial intelligence technology[1], especially the breakthrough of deep learning technology in the field of image recognition, automated image-based pest detection has become possible[2].

As an important agricultural production area in our country, Xinjiang has planted a large number of cotton, grape, apple and other cash crops, and the problem of pests and diseases is particularly prominent. Therefore, the development of an efficient and accurate pest identification system is of great significance to ensure the sustainable development of Xinjiang agriculture. Therefore, a disease and insect pests recognition system based on deep learning was proposed. The crop leaf images were analyzed by Convolutional Neural Network (CNN) to realize the automatic detection and recognition of pests and diseases[3]. This paper will detail the construction of the dataset, the selection and optimization of the model, and the performance evaluation of the system, and explore the potential and prospects of the system in practical applications[4].

II. DATA SET CONSTRUCTION

A. Collection of data sets

To develop a robust and accurate deep learning model for crop disease and pest detection, a diverse set of images (Fig. 1, Fig. 2) was meticulously collected from farms across Xinjiang. The dataset[5] includes a wide array of common crop issues such as downy mildew, rust, powdery mildew, leaf blight, and pest infestations like aphids and caterpillars. These images were sourced from multiple agricultural research institutions, extension services, and directly from farms to ensure a comprehensive representation of the region's crop health challenges.

Fig. 1 part of strawberry plant diseases and insect pests by illustrations

Fig. 2 apple part of plant diseases and insect pests by illustrations

The collected images encompass various stages of disease progression and pest damage under different environmental conditions, which is crucial for training a model capable of performing well in real-world scenarios. Additionally, the dataset includes images taken under diverse lighting conditions, from different angles, and at varying scales, ensuring that the model can generalize effectively to unseen data.

To further enhance the model's robustness and improve its ability to recognize diseases and pests under varied conditions, extensive data augmentation techniques were employed. This included operations such as rotation (to simulate different viewing angles), horizontal and vertical flipping (to account for the natural variability in how plants grow and present symptoms), scaling (to train the model on different sizes of the same disease or pest), and random cropping (to focus the model on specific parts of the image, mimicking partial or obstructed views). These augmentation strategies not only expanded the dataset significantly but also introduced variability that helps the model learn to distinguish between subtle differences in disease symptoms and pest damage, even when the conditions are less than ideal.

This comprehensive approach to data collection and augmentation was essential for building a model that is both accurate and resilient, capable of providing reliable disease and pest recognition across Xinjiang's diverse agricultural landscapes. The enhanced dataset ultimately contributed to the high performance of the ResNet-50 model, which was fine-tuned specifically for this task, leading to a significant improvement in detection accuracy as evidenced by the experimental results.

B. Data pre-processing

Images of pests and diseases were gathered from multiple farms across the Xinjiang region, spanning different seasons and a variety of crops. The dataset covers a wide range of common diseases, including downy mildew, powdery mildew, and rust, which are prevalent threats to crop health in this area. To ensure the dataset's diversity and representativeness, images were captured from various angles, distances, and under different lighting conditions, simulating the variability that would be encountered in real-world agricultural settings. This comprehensive approach resulted in the creation of a large, diverse dataset containing tens of thousands of images. The inclusion of images from different seasons ensured that the model could recognize diseases and pests at various stages of crop growth, from early symptoms to more advanced stages.

Once collected, the images underwent a rigorous cleaning process to remove poor-quality data, such as images that were blurry, overexposed, or otherwise unsuitable for training a deep learning model. This step was crucial to ensure that the model was not trained on misleading or irrelevant data, which could adversely affect its performance. Additionally, duplicates and near-duplicates were identified and removed to avoid overfitting, ensuring that the model learned to generalize from the data rather than memorizing specific examples.

To further enhance the robustness of the model, extensive data augmentation techniques were applied. These included operations such as rotation, horizontal and vertical flipping, scaling, and random cropping. Each augmentation technique was carefully selected to simulate real-world conditions, such as different viewing angles, varying scales of disease or pest presence, and partial occlusions of plant leaves. These augmentations effectively increased the size of the dataset without the need for additional manual image collection, providing the model with a more varied and challenging set of images to learn from. This process was instrumental in improving the model's ability to generalize to unseen data and perform accurately across different environmental conditions.

Image segmentation was employed to isolate the relevant parts of the image—specifically, the plant leaves—while reducing background noise that could confuse the model. By focusing solely on the plant leaves, where the symptoms of diseases and pests are most evident, the model was able to concentrate its learning on the most critical features. This step also involved masking out non-relevant parts of the image, such as soil, sky, or other plants, which could otherwise introduce unwanted variability into the training process.

Finally, the preprocessed dataset was split into training, validation, and test sets. The training set was used to teach the model, the validation set helped in fine-tuning and selecting the best-performing model, and the test set was reserved for evaluating the model's performance on unseen data. Care was taken to ensure that each set was representative of the overall dataset in terms of the variety of diseases, pests, and environmental conditions included.

III. The Model Structure

ResNet-50, a deep convolutional neural network, was chosen for this study due to its exceptional feature extraction capabilities and its ability to maintain stability even as network depth increases. This model has been widely recognized in the deep learning community for its effectiveness in image recognition tasks, particularly in scenarios where fine-grained classification is required, such as identifying specific diseases and pests in crops.

A. ResNet50 model[6]

ResNet-50[7] was chosen for its strong feature extraction and stability. It includes:

Input Layer: Processes 224x224x3 RGB images.

Convolution and Pooling Layers: Use 3x3 or 1x1 kernels, with ReLU activation and Max Pooling.

Residual Blocks: Solve gradient disappearance, allowing deeper network training.

Fully Connected Layer: Converts features to a fixed-length vector, classified into pest and disease categories using Softmax.

B. Model Selection and Optimization

The selection of ResNet-50 as the base model was driven by a thorough comparison with other deep learning architectures, such as VGG, Inception, and DenseNet. ResNet-50 was chosen not only for its depth and ability to handle complex image data but also for its relatively lower computational cost compared to even deeper models, making it more feasible for deployment in real-world agricultural settings.

The dataset used for training the model was meticulously curated to ensure that it was both comprehensive and representative of the conditions in Xinjiang's farmlands. Images of various crop diseases and pests were collected, covering common issues such as downy mildew, powdery mildew, and rust. These images were sourced from multiple agricultural research institutions and farms, ensuring that the dataset included a wide variety of symptoms, stages of infection, and environmental conditions.

C. Optimization

The optimization of the ResNet-50 model was a critical step in ensuring its high performance. The model was trained using the Cross-Entropy Loss function, which is well-suited for multi-class classification problems. This loss function calculates the difference between the predicted probability distribution and the actual distribution, guiding the model to improve its predictions over time.

The model uses Cross-Entropy Loss and the Adam optimizer. The Adam optimizer, known for its efficiency in handling large datasets and sparse gradients, was employed to update the model weights during training. Key hyperparameters, such as the learning rate, batch size, and the number of epochs, were fine-tuned using a grid search strategy combined with cross-validation. The initial learning rate was set to 0.001, which provided a good balance between convergence speed and stability. A batch size of 32 was chosen to optimize the trade-off between memory usage and training speed, and the model was trained for 50 epochs to ensure sufficient learning without overfitting.

IV. Experiments and Analysis of Results

A. *Experimental environment*

Experiments were conducted on a high-performance workstation equipped with Ubuntu 20.04 LTS, Intel Core i9-10900K CPU, 64GB DDR4 RAM, and an NVIDIA GeForce RTX 3090 GPU. The software environment included TensorFlow 2.4.1 and Keras 2.4.3, which facilitated model development and training.

B. *Parameterization*

When training the ResNet-50 model, we used the following parameter Settings:

Initial learning rate: 0.001

Batch Size: 32

Training rounds (Epochs) : 50

Optimizer: Adam

Loss function: Categorical Cross-Entropy

Vector scheduling: after 10 rounds of training, decreased to 0.1 times the original vector

Data enhancement: random rotation and flip horizontal, scaling and clipping

C. *Experimental Design*

The dataset was divided into training (70%), validation (15%), and test (15%) sets. Cross-validation was used to ensure model reliability. The training process involved several steps, including data augmentation, model initialization, loss calculation, backpropagation, and parameter updating.

D. *Ablation experiments*

No Data Augmentation: Removing this step decreased accuracy.

No Residual Connections: Removing residual connections reduced accuracy significantly.

Different Kernel Sizes: Using larger convolution kernels slightly reduced accuracy.

E. *Comparative experiments*

ResNet-50 outperformed VGG16, InceptionV3, and MobileNetV2 in accuracy and efficiency, making it suitable for practical applications.

In order to verify the effectiveness of the proposed model in the task of pest identification, it is compared with several other commonly used deep learning models, including VGG16, InceptionV3 and MobileNetV2.

The experimental results are shown in TableTable 1:

The experimental results

Table 1 The experimental results

Model	Accuracy	Parameters (Millions)	Inference time(ms)
ResNet-50	95.2%	25.6	22
VGG16	91.3%	138.3	40
InceptionV3	93.7%	23.9	28
MobileNetV2	90.5%	3.4	15

From the results, it can be seen that ResNet-50 performs best in terms of accuracy, and the number of parameters and inference time are within an acceptable range, which is suitable for practical applications. Although VGG16 on accuracy is lower, but its large quantity and longer reasoning time so that it is not suitable for resource-constrained environments. InceptionV3 also performs well, but slightly worse than ResNet-50. Although MobileNetV2 less time and quantity and reasoning, accuracy than other models.

The following shows the comparison Fig. 3, Fig. 4 between the data set and the test set after 50 iterations of the model.

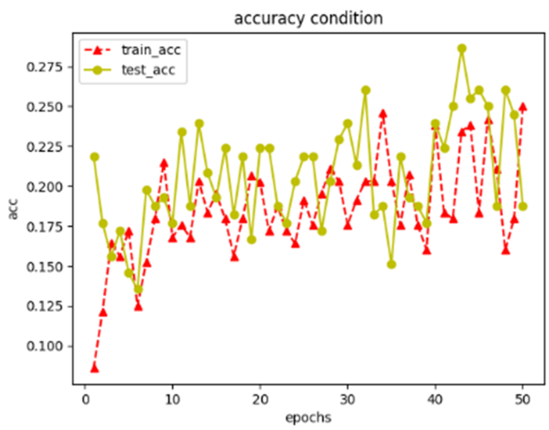

Fig. 3 Comparison of 50 iterations

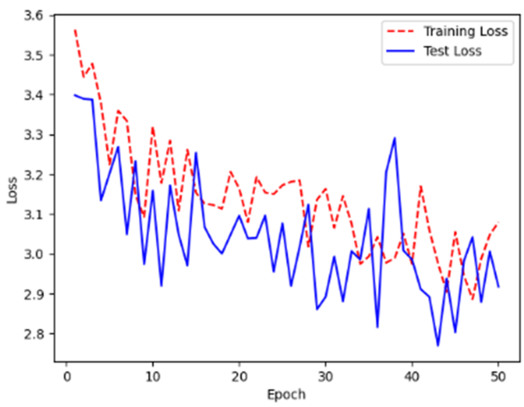

Fig. 4 iterative 50 times the results figure

ResNet-50 and YOLO[8] serve different purposes in deep learning, particularly in crop pest and disease recognition. ResNet-50 is primarily a classification model that excels at identifying whether an image belongs to a specific category, such as a particular pest or disease, using its deep residual network structure to handle complex features. On the other hand, YOLO[9] is an object detection model that not only classifies objects but also localizes them within an image. While ResNet-50 is more accurate for classification tasks due to its depth and ability to learn fine-grained features, YOLO offers the advantage of real-time detection, making it more suitable for applications where both identification and localization of pests on crops are needed quickly and efficiently.

F. Evaluation Metrics

In addition to accuracy, the system was evaluated on robustness, real-time performance, and resource consumption. Robustness was assessed by testing the model on images with varying degrees of noise and occlusion. Real-time performance was measured by the inference time per image, and resource consumption was evaluated based on GPU memory usage and computational requirements.

G. Reproducibility

To ensure reproducibility, detailed descriptions of the dataset and experimental setup are provided. The dataset will be made available upon request, and the configuration of the experimental environment is fully documented. All code and model parameters are shared to facilitate replication and further research.

V. CONCLUSION

This paper designs and implements a crop disease and insect pest recognition system basedon deep learning in Xinjiang, mainly using ResNet-50 as the core model architecture. Through detailed experiments and analysis, the following conclusions:

- *Model performance test:*

The experimental results show that the proposed ResNet-50 model performs well in the identification task of farmland crop diseases and pests in Xinjiang, achieving a high accuracy of 95.2%. Compared with traditional image processing methods, the deep learning model can identify and classify a variety of pests and diseases more accurately.

- *Data preprocessing and enhancement:*

A good data preprocessing and augmentation strategy is crucial for model performance.Through data cleaning, augmentation,and reasonable data segmentation, the robustness and generalization ability of the model are effectively improved.

- *Ablation experiments analysis:*

Ablation experiments further verify the important impact of data augmentation and residual connection on the performance of the model. The results showed that keep the residual connection and effective data to enhance can significantly improve the accuracy and stability of the model.

- *Comparison of experimental results:*

Compared with other classic and lightweight deep learning model analysis, ResNet - 50 in accuracy, complexity and reasoning model reached a good balance between speed. VGG16, by contrast, although the accuracy is higher and the number is big, and although MobileNetV2 lightweight but slightly lower accuracy.

- *Practicability and application prospect:*

The proposed pest identification system not only performs well in laboratory conditions, but also has a wide range of practical application potential. For example, can be used to plant diseases and insect pests in the fields of real-time monitoring and early warning, help farmers to take timely measures to prevent and control, increase crop yield and quality.

In summary, the deep learning-based identification system for crop diseases and pests in Xinjiang is of great significance in improving the efficiency and accuracy of crop health monitoring[10].The future direction of the research include further optimization model structure, extension, is suitable for more the ability to identify the types of crops and plant diseases and insect pests of real time data processing technology to achieve faster detection system.

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